Denis Marfin

Data Analyst | BI and Product Analytics | SQL, Python, DWH

Belgrade, Serbia | heissassss@gmail.com | Remote or hybrid

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Summary

Data Analyst with 3+ years of experience in banking and the public sector, covering BI, data analysis and product analytics (funnels, cohorts, A/B tests). Strong hands-on SQL (PostgreSQL, MS SQL / T-SQL) and Python (Pandas, NumPy) for data extraction, validation and reporting automation, plus dashboards in Power BI (DAX, Power Query) and Apache Superset. Investigates data discrepancies down to the root cause, builds data quality controls and prepares management reporting for senior stakeholders, including members of a bank's Management Board. Master's student in Data Analysis and Artificial Intelligence.

Core Skills

SQL: PostgreSQL, MS SQL / T-SQL, window functions, CTEs, query optimisation, indexes, data marts, complex queries on large datasets

Python: Pandas, NumPy, openpyxl, Matplotlib, Plotly, Jupyter, automation of data extraction, processing, validation and reporting

BI and Visualization: Power BI (DAX, Power Query), Apache Superset, Tableau, self-service dashboards, MS Excel (advanced), VBA

Product and Business Analytics: funnel analysis, cohort analysis, A/B testing, hypothesis testing, exploratory data analysis (EDA), KPI and metric definition, ad hoc analysis, management reporting

DWH and Data Quality: DWH-based analytics, ETL, data migration, source-to-target mapping, AS IS / TO BE analysis, reconciliation, root cause analysis, data completeness and consistency monitoring

Machine Learning (working knowledge): scikit-learn, TensorFlow / Keras, CNNs, model evaluation (accuracy, F1)

Tools: Git, Docker, Jira, Confluence, Airflow (basics), Azure / GCP (basics)

Professional Experience

Data Analyst, VTB Bank (PJSC) | Sep 2024 – 2026
Digital Operating Model Development Department

- Built an automated workforce analytics model (PostgreSQL, Excel Power Query) covering about 50,000 IT employees, 2,000+ information systems and 18 production roles, split by category (staff, contractor, bank employee) in absolute and percentage terms; cut report preparation from 8 hours to 5 minutes.
- Showed top management which systems depend on contractor staff; the analysis became the basis for a process to control workforce shares and move key IS roles in-house.
- Carried out the migration of BI reporting from Power BI to Apache Superset: audited 40+ dashboards, ranked them by business criticality, proposed merges and retired 55%; migrated the rest with no loss of business logic or metrics, connected the database and revoked legacy access.
- Fixed a dashboard query that repeatedly overloaded the database by rewriting it with indexes and reducing the data volume it pulled; load-tested all migrated dashboards.
- Built a data quality control system in Python (Pandas, NumPy) monitoring completeness, accuracy and consistency across about 30 sources; the report generator with a desktop interface cut a regular data quality report from one working day to 30 seconds and delivers a separate file to each of 11 departments.
- Performed AS IS / TO BE analysis of reporting, investigated discrepancies between data sources down to the root cause and reduced data deviations by 20%.
- Took part in data migration projects: source analysis, entity mapping, validation of transfer correctness.
- Wrote SQL (PostgreSQL, MS SQL) for analytical extracts, data marts and intermediate calculation layers; designed and maintained dashboards and self-service data marts in Power BI and Apache Superset.
- Tested business hypotheses with A/B tests and cohort analysis; documented calculation logic for developers and business users.
- Defined and maintained KPI calculation logic, delivered 200+ ad hoc analytical requests per year and prepared management reports for members of the Management Board.

Analyst, Ministry of Culture of the Russian Federation | Aug 2023 – Nov 2024

- Consolidated data from multiple sources and formats into a single standard structure and analysed it to prepare management reports.
- Optimised the reporting process, reducing the workload of the departments involved.
- Coordinated data collection and data quality issues across departments.

Projects

- Affiliate Funnel Dashboard (Power BI, DAX, Power Query): player acquisition funnel by traffic type, affiliate performance and cohort conversions, with 10 documented data quality findings. github.com/heissasss/PowerBI_dash
- CIFAR-10 CNN Classifier (TensorFlow, Keras, Docker): custom CNN built from scratch, architectures compared on accuracy, F1 and FPS, Gradio demo. github.com/heissasss/classification
- Master's thesis: information and analytical system for investment asset management using machine learning.
- Academic publications on LLM evaluation in investment analysis and on ensemble models (FinBERT with quantitative models).

Education

Master's degree, Data Analysis and Artificial Intelligence
RANEPA, Institute of Economics, Mathematics and Information Technologies | 2025 – 2027 (in progress)

Bachelor's degree, Applied Informatics
State University of Management, Institute of Information Systems | 2021 – 2025

Additional Training

- MTS Data Analytics School (Teta): ML Engineer / Data Scientist programme, 2025
- Karpov.Courses: Data-Driven Decision Making and AI

Languages

Russian: native | English: B2 (Upper-Intermediate), business level, spoken and written